

- 7** Increased levels of atmospheric carbon dioxide can lead to an increased rate of photosynthesis. This is called the CO₂ fertilisation effect.

This effect varies depending on air and soil temperature, the availability of water, the availability of nutrients and the species of plant.

Plants that can take advantage of higher carbon dioxide levels will have a greater increase in biomass.

- (a) Describe how carbon becomes incorporated into plant biomass.

(3)

- (b) (i) Explain why air temperature affects the CO₂ fertilisation effect.

(2)

(ii) Suggest why soil temperature affects the CO₂ fertilisation effect.

(2)

(c) Explain why a lack of water availability in the leaves would decrease the CO₂ fertilisation effect.

(3)

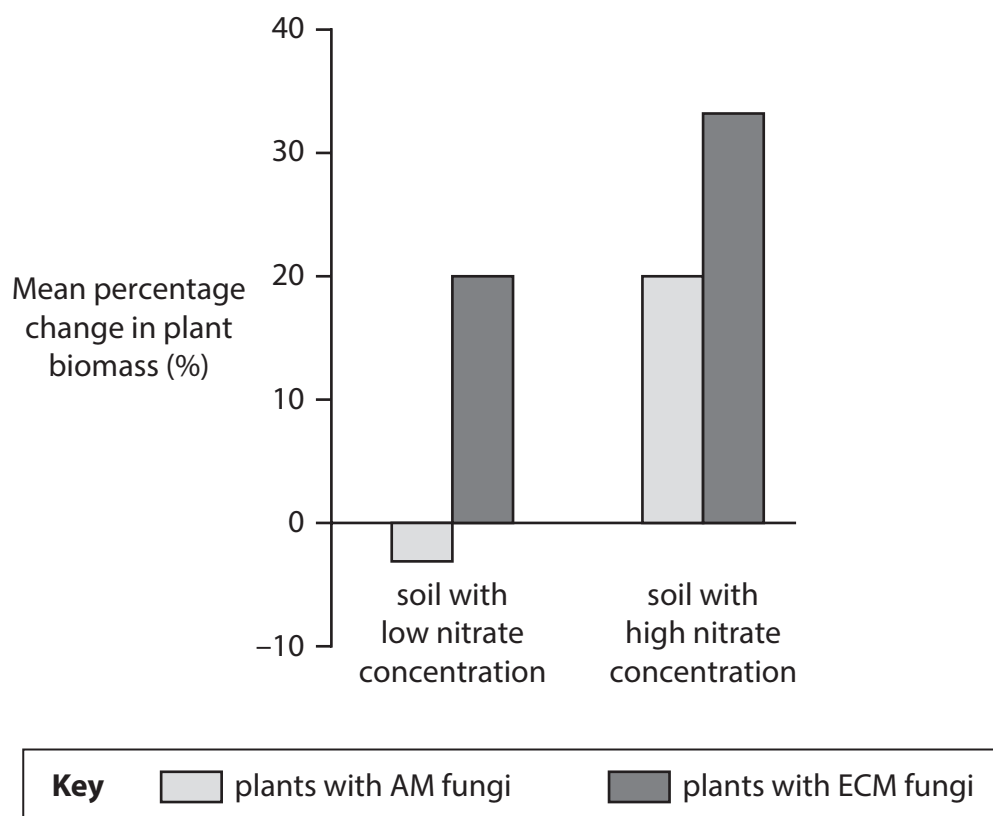
- (d) The roots of most land plants are associated with microscopic fungi called mycorrhizal fungi.

These fungi provide the plants with nitrate ions.

The effect of two types of mycorrhizal fungi, ECM and AM, on the change in plant biomass was investigated.

Plants were grown in soil containing either low or high concentrations of nitrate.

The graph shows the results of this investigation.



Explain the results of this investigation.

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(Total for Question 7 = 14 marks)